Group Theory Notes

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Outline

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Introduction

Examples of Groups

Rubik's cube group

Permutations

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Rubik's Cube

Example (Rubik's cube)

Consider the Rubik's cube (for an online example, see here). What moves can we perform on a Rubik's cube?

Rules of moves

There are many, *many* different configurations (or states) possible, but only one solution. We can get from one state to another by performing a valid movement of the cube. These movements have the following properties:

1. There is an unchanging list of allowable moves.
2. Every move is reversible.
3. Every move is deterministic.
4. Moves can be combined.

Rubik's Cube
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Example (Generators of Rubik's cube motions)

The website above uses twelve basic moves: F, F′, U, U′, and so on.

- How can you write F′ in terms of F?
- What is the smallest number of moves you can find that represent all of the basic moves on a Rubik's cube?

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Groups

The observations above form the basis of a mathematical structure called a group. Groups appear throughout mathematics and its applications.

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Examples of Groups

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Example (Coins in a line)

Consider placing a penny and a nickel side by side in front of you. Is the action of swapping the positions of both coins a group? Note that you will need to specify an appropriate list of moves and determine if that list satisfies the above rules.

Example (Marbles in piles)

Suppose you have five marbles in each of two piles in front of you. You have two actions available to you: you can take a marble from the left pile and move it to the right pile, *or* you can take a marble from the right pile and add it to the left pile. Is this a group?

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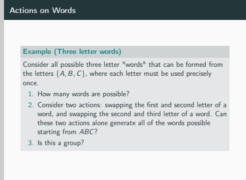
Example (Three letter words)

Consider all possible three letter "words" that can be formed from the letters $\{A, B, C\}$, where each letter must be used precisely once.

1. How many words are possible?
2. Consider two actions: swapping the first and second letter of a word, and swapping the second and third letter of a word. Can these two actions alone generate all of the words possible starting from ABC?
3. Is this a group?



End of class *<2026-06-22 Mon>*

Group Theory

At this point, you might start thinking of a group as a set of objects that we can move around in a nice way. Group theory is about the study of such sets following the rules outlined above.

Example (Coins in a line (again))

Consider the group of coins in a line.

1. By rule 4, the action from this exercise when performed twice in a row yields another action in the group. What is this action?
2. Does every group have an action like this?

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As a reminder, the group rules are as follows:

1. There is an unchanging list of moves.
2. Every move is reversible.
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Example (Symmetry groups)

Describe the symmetry group of an equilateral triangle and a non-square rectangle.

Examples of Groups
Example (Symmetry groups)
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Recall that the symmetry group of a shape is the set of reflections/rotations that leaves the shape unchanged.

Groups with Certain Properties

Example (Groups with restrictions)

Try to find or construct groups that satisfy the following constraints:

1. The order in which the actions are performed matters.
2. The order in which the actions are performed doesn't matter.
3. There are exactly three actions.
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1. Rubik's cube group or symmetry group of triangle/square
2. Symmetry group of non-square rectangle or coin group
3. Look at triangle
4. Look at square or non-square rectangle
5. Can use the integers or an infinite chain

Nonexamples of Groups

Example (Sets that break group rules)

Devise a situation that satisfies all of the group rules except for the second, third or fourth depending on your group.

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Devise a situation that satisfies all of the group rules except for the second, third or fourth depending on your group.

Break class into three groups (count off 1, 2, 3) and have the groups focus on breaking a single rule from group rules.

1. Breaks second rule: can use naturals with addition or an action that "erases" information.
2. Think of symmetry group where we rotate or flip based on results of coin toss.
3. We already have an example of this. We can also look at limited symmetry groups (i.e., subgroups that aren't closed).

Rubik's cube group

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Terminology

The cubies on a Rubik's cube can be placed in the following categories:

1. Corner cubies
2. Edge cubies
3. Center cubies

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Cubies and Cubicles

- A cubicle is the position of a given cubie. Moves changes the positions of the cubies, but *not* the cubicles.
- Positions can be specified by listing incident faces: ufr represents the cubicle/cubie on the upper front right face (i.e., a corner).

Example (Edge Positions)

Give all positions for edge cubies on the back face of a cube.

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Group Actions and Cubies

Example (Generators Acting on Cubies)

What happens to cubies in the previous categories if you apply any of the six basic Rubik's cube actions?

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Example (Size of the Rubik's cube group)

The Rubik's cube group is based on the six fundamental motions F, B, U, D, L and R. How large could this group be? *Hint*: try answering the following questions.

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Estimating Rubik's Cube Group
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1. The corner cubies can be arranged in up to 8! ways with up to 3 different configurations each, giving an upper bound of $8!3^8$ arrangements for the corner cubies.
2. The edge cubies have an upper bound of $12!2^{12}$ arrangements.
3. The upper bound for moves is therefore $8!3^8 12!2^{12}$.

Permutations

Permutations

Permutation

- A permutation of an ordered collection of objects is a rearrangement of those objects.
- Every group action we've looked at so far is an example of a permutation.

Example (Permutations and groups)

List the permutations and the objects being permuted for the following groups:

1. The three letter word group (see Example (Three letter words)).
2. The symmetry group of the equilateral triangle (see Symmetry Groups (slide)).

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Symmetry Groups

- Shapes have symmetry groups (also called *dihedral groups*).
- Collections of objects also have symmetry groups. The actions are permutations.

Example (Elements of a symmetry group)

Give two elements of the symmetry group of $\{1, 2, 3, 4\}$.

Example (Number of elements of a symmetry group)

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Note the connection to previous permutations!

S_n

- The symmetry group on n elements is called S_n.
- You've already found S_3!
- Permutations in S_n can be written in cycle notation.

2026-06-23

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└Permutations
└S_n

S_n

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Cycle Notation

The permutation $\sigma \in S_3$ given by

$$\begin{pmatrix} 1 & 2 & 3 \\ \downarrow & \downarrow & \downarrow \\ 2 & 3 & 1 \end{pmatrix}$$

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Example (Cycle notation in S_4)

Write the permutations you found in Example (Elements of a symmetry group) using cycle notation.

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Combining Permutations

- Cycle notation provides a convenient way to combine permutations.

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